

Unit 14, Lesson H1: Creating Relative Frequency Tables



Benchmarks

MA.912.DP.3.2 Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical and whether it is univariate or bivariate, and interpret the different components and quantities in the display.



Learning Target

☐ I can construct a two-way relative frequency table.



14.H1.1 Warm-Up: What Percentage?

1. Mindy is in charge of ordering supplies for the break room at her workplace, so she surveys her coworkers. She asks them if they prefer coffee or tea and whether or not they like sugar in their beverages. Her findings are shown in the two-way table.

	Coffee	Tea	Total
Sugar	15	8	23
No Sugar	25	2	27
Total	40	10	50

Use the two-way table to answer the following questions.

- a. What percentage of people surveyed prefer coffee?
- b. What percentage of people surveyed drink tea with sugar?



14.H1.2 Exploration: Two-Way Relative Frequency Tables

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology. 741 teens who use social media were asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral) and how much time they believe they spend with their friends in person (too much, too little, or about the right amount). The results are summarized in the joint frequency table.

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

The data can also be displayed in a joint relative frequency table, where the cells represent the joint and marginal relative frequencies. These are often useful to construct when given percentages for a data set, rather than frequencies.

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	2.97%	2.83%	2.7%	8.5%
	Too Little	11.74%	10.39%	14.17%	36.3%
	Just Right	14.84%	15.25%	25.10%	55.20%
	Total	29.55%	28.48%	41.97%	100.00%



A **joint relative frequency** is the ratio of a joint frequency and the total number of data points.

A **marginal relative frequency** is the ratio of a marginal frequency and the total number of data points.

1. What is the sum of the joint relative frequencies?
2. What is the sum of the marginal relative frequencies for the variable "effect on people your age?"
3. What is the sum of the joint relative frequencies for those who believe the effect of social media is mostly positive?

4. What is the sum of the joint relative frequencies for those who believe they spend too much time with friends in person?

5. Use only the values from the joint frequency table to answer the following questions.
 - a. What percentage of those who spend too much time with friends in person believe that the effect of social media has been mostly positive?

 - b. What percentage of those who spend too much time with friends in person believe that the effect of social media has been mostly negative?

The percentages calculated in Parts A and B above are **conditional relative frequencies**.



A **conditional relative frequency** is the ratio of a joint relative frequency to a marginal relative frequency. Equivalently, the ratio of a joint frequency and a marginal frequency.



14.H1.3 Guided Instruction: Creating a Two-Way Relative Frequency Table

Consider the following data on whether or not animals received a microchip from the Kirkman Road Veterinary Clinic's database of canines (dogs) and felines (cats).

- 37.61% of animals in the sample were felines.
- 30.63% of animals in the sample did not have microchips.
- Of the animals with microchips, 81.17% were canines.
- 34.73% of the felines had microchips.

1. Complete each statement.

a. 37.61% represents a frequency.

- ☐ joint
☐ marginal
☐ conditional

relative

b. 30.63% represents a frequency.

- ☐ joint
☐ marginal
☐ conditional

relative

c. 81.17% represents a frequency.

- ☐ joint
☐ marginal
☐ conditional

relative

d. 34.73% represents a frequency.

- ☐ joint
☐ marginal
☐ conditional

relative

- joint
- marginal
- conditional

	Microchipped	Not Microchipped	Total
Feline			
Canine			
Total			

- felines.
- canines.

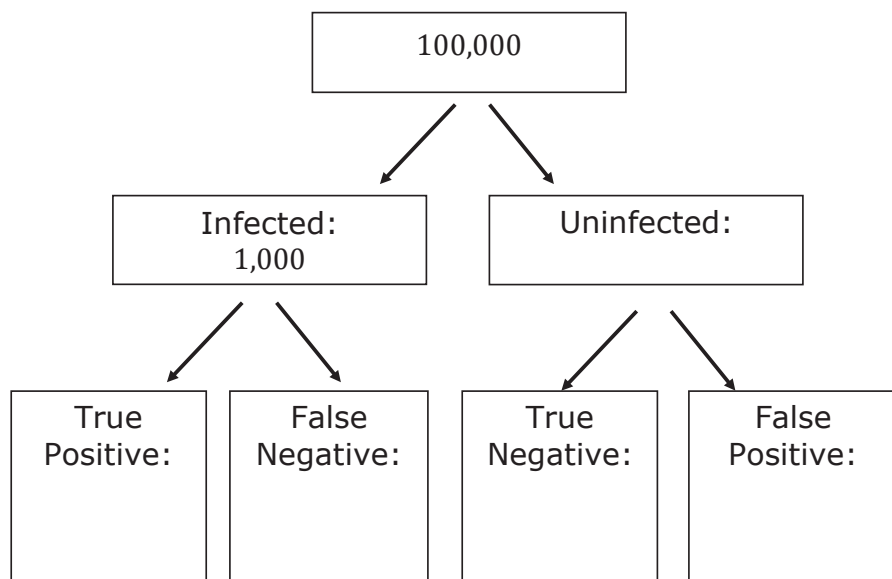
7. Which kind of pet is more likely to be microchipped? Explain your reasoning.



14.H1.4 Guided Practice: Creating a Two-Way Relative Frequency Table

A study is performed to determine the efficacy of a new test that determines whether or not a patient has a disease. 100,000 people were included in the study. 1% of the participants are infected with the disease. Of those infected, the test correctly identified the presence of the disease in 750 patients, and 250 patients received a false negative. For those patients that were not infected with the disease, the test was correct in 98,505 cases.

1. Use this data to complete the tree diagram.



2. Complete the two-way relative frequency table to represent the data.

	Positive	Negative	Total
Infected			1%
Not Infected			99%
Total			



14.H1.5 Try It: RMS Titanic

- The RMS Titanic was a British passenger liner operated by the White Star Line that sank in the North Atlantic Ocean on April 15, 1912, after striking an iceberg during her maiden voyage from Southampton to New York City. Of the estimated 2,201 passengers and crew aboard, 1,490 died, making the sinking, at the time, one of the deadliest of a single ship.

Passengers can be classified as whether the person survived (Yes or No) and the class of the ticket they had (1st, 2nd, 3rd, or Crew).

- 14.8% of passengers had 1st class tickets.
- 13.0% of passengers had 2nd class tickets.
- 35.4% of those who died were third class passengers.
- 25.0% of those who survived were 3rd class passengers.
- 76.0% of the crew died.
- 62.5% of 1st class passengers survived.

Create a two-way relative frequency table based on the data provided.

Ticket	Survived		Total
	No	Yes	
1st			
2nd			
3rd			
Crew			
Total			



14.H1.6 Wrap-Up: Reflection

- Complete each statement.
 - After today's lesson, I am still unsure about ...
 - To fill this gap, I intend to ...